

Whole program: what each optimization bought, per program.

Vanilla Gibbon is AoS, recursive traversal, immutable cursors. Each later cell is what that optimization alone bought over the column to its left, so a row multiplies to its total.

Colour is clipped at $4\times$ either way; an outlined cell is beyond that, and its number rather than its shade carries the value.

Compiler	1×	1.03×	1.2×	1.17×	3.23×	0.973×	0.95×	3.74×	1×	0.99×	16.1×
reduceNestedList	1×	0.996×	1.78×	1.05×	7.52×	1×	0.995×	1×	1×	1.01×	14.1×
LinearListReduction	1×	1.01×	1.67×	1.04×	5.99×	1.01×	0.588×	1.03×	0.997×	1×	6.38×
Add1TreeInt8	1×	1.01×	1.7×	1.01×	1.15×	0.987×	1.94×	1.37×	1.09×	1×	5.71×
Add1TreeInt16	1×	1.01×	1.77×	0.941×	1.16×	0.99×	1.85×	1.35×	1.03×	1.03×	5.11×
PiecewiseFunctions_addConstPW	1×	1.01×	1.05×	0.887×	1.04×	0.968×	1.22×	4.3×	1.03×	0.997×	5.08×
ArithmeticIntensityInt8	1×	1×	1.23×	0.982×	1.05×	0.813×	1.06×	1.04×	4.14×	1×	4.73×
DecisionTree	1×	0.976×	2.04×	1.33×	1.67×	1.01×	0.994×	0.997×	1.01×	0.991×	4.4×
Add1TreeInt32	1×	1×	1.77×	0.94×	1.12×	0.991×	1.63×	1.33×	1×	1.02×	4.11×
Trie	1×	1.01×	1.27×	0.991×	1.08×	1.03×	0.974×	2.71×	0.983×	1.01×	3.69×
OctTree_clearFlags	1×	1×	1.01×	0.682×	1.02×	0.992×	1.18×	4.52×	0.99×	1×	3.66×
ArithmeticIntensityInt16	1×	1×	1.42×	0.952×	1.14×	0.423×	1.06×	1.02×	5.09×	1.02×	3.66×
MonoTree	1×	0.988×	1.64×	1.06×	1.12×	1×	1.42×	1.16×	1.01×	0.997×	3.22×
Add1TreeInt64	1×	0.997×	1.59×	0.95×	1.12×	1.01×	1.44×	1.27×	1.01×	0.946×	2.99×
PiecewiseFunctions_diffPW	1×	0.994×	1.03×	0.864×	1.01×	1×	1.13×	2.77×	1.04×	1.02×	2.94×
ObjectGraph	1×	1.01×	1.57×	0.921×	1.16×	1.15×	1.15×	1.24×	1.06×	0.985×	2.91×
DBQuery	1×	0.984×	1.23×	0.903×	1.11×	1.01×	1.05×	2.25×	1.01×	0.988×	2.9×
ArithmeticIntensityInt32	1×	0.99×	1.29×	0.978×	1.08×	0.694×	1.06×	1.03×	2.71×	1.01×	2.81×
PiecewiseFunctions_truncateTolViolations	1×	0.992×	1.7×	1.27×	1.36×	1.01×	0.756×	0.999×	0.999×	1×	2.22×
PiecewiseFunctions_compressMass	1×	1×	1.65×	1.51×	1.18×	0.991×	0.761×	1×	0.999×	1×	2.21×
OctTree_scaleEnergy	1×	1.05×	1×	0.657×	1.03×	0.993×	1.15×	2.51×	1.01×	1.03×	2.13×
DomTree	1×	1.03×	1.52×	1.26×	1.05×	0.991×	0.869×	1.18×	1×	1.01×	2.1×
PiecewiseFunctions_pmapCutHistogram	1×	1×	1.89×	0.922×	1.7×	0.85×	0.909×	0.628×	1.7×	0.85×	2.09×
PiecewiseFunctions_norm2Estimate	1×	1×	2.13×	0.717×	1.5×	1.09×	0.827×	1×	1.05×	0.913×	1.98×
PiecewiseFunctions_lbDeuxLoadProxy	1×	0.436×	3.88×	1.13×	1.15×	1.03×	0.799×	0.996×	0.999×	0.978×	1.77×
TernaryTree	1×	0.904×	1.43×	0.932×	1.01×	1.01×	1.31×	1.09×	0.966×	1.02×	1.73×
PiecewiseFunctions_autorefineMaxLevel	1×	0.998×	1.59×	1.2×	1.04×	1.03×	0.814×	0.996×	0.969×	0.977×	1.57×
ArithmeticIntensityInt64	1×	0.998×	1.19×	1.01×	1.04×	0.996×	1.01×	1.04×	1×	1×	1.31×
KDTree	1×	0.956×	1.41×	1.11×	1.02×	1.18×	0.94×	0.999×	1.01×	0.724×	1.24×
DecisionTreeClassify	1×	0.902×	1.01×	0.86×	1.24×	1.05×	1.01×	0.991×	1×	0.971×	0.984×
OctTree_sumMass	1×	0.917×	1.32×	0.701×	1.13×	0.988×	0.928×	1×	1.01×	0.977×	0.863×
OctTree_sumEnergy	1×	0.989×	1.27×	0.706×	1.05×	1.02×	0.925×	0.996×	1.01×	0.976×	0.853×
OctTree_countParticles	1×	1.02×	1.18×	0.732×	1×	0.981×	0.93×	1.01×	1.01×	0.993×	0.812×
OctTree_fmmPotential	1×	0.962×	1.04×	0.841×	0.991×	1.01×	0.965×	0.999×	0.994×	0.994×	0.801×
OctTree_barnesHutPotential	1×	0.996×	1.31×	0.704×	0.947×	1.05×	0.942×	0.992×	1×	0.934×	0.799×
OctTree_countActive	1×	1.01×	1.18×	0.735×	0.976×	1×	0.925×	1.01×	0.97×	1.03×	0.798×
ColorOctree	1×	1.01×	1.26×	0.738×	0.981×	1.1×	0.887×	1.01×	1.01×	0.466×	0.429×
Vanilla GibbonC tail calls offMutable cursorsSoA layoutC tail calls onC auto-vec. offLoopificationBuffer sharingGibbon SIMD C auto-vec. onTotal											